

IMprecision is IMperative for reliable UQ (and it's not IMpossible with possibilistic IMs)

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- UQ can mean different things:
 - belief assignments, e.g., probabilities
 - confidence sets
 - ...
- It's concerning that we have fundamentally distinct & acceptable approaches that tend to give different answers
- We shouldn't have to choose!
- *Inferential models* (IMs) offer “probability-like” output for UQ that's frequentistly reliable
- My “probabilities” are *imprecise probabilities*, but you'll see that this isn't so unfamiliar

- Key observation: *imprecision is imperative!*
- That is, basic reliability properties fail when UQ is precise²
- But not just any imprecise probability can fix this
- Special *possibilistic* form is strongly suggested by the reliability objectives themselves
- New *possibilistic IM* solution seems quite promising³

IMs remind me of conformal prediction. —Vovk (to me, 2019)

²Related to the *false confidence theorem* in Balch et al (*PRSA* 2019)

³M. (*JASA* 202x, [arXiv:2507.09007](#))

- Problem setup
- Reliability and its *imprecision-is-imperative* consequences
 - belief assignments
 - confidence sets
- Possibility theory and possibilistic IMs
 - probability-to-possibility transform
 - “principle of minimum complexity” → conditioning
- Connection to conformal prediction
- Conclusion

Problem setup

- Data $Z^n \in \mathbb{Z}^n$, observed value z^n
- ML applications: often $z = (x, y)$ is a feature–label pair
- Joint probability distribution P for Z^n , unknown
- Uncertain quantity of interest:
 - a new observation Z_{n+1}
 - a functional $\Theta = \phi(P)$, e.g., risk minimizer
- **Goal:** *quantify uncertainty about Z_{n+1} , given $Z^n = z^n$*

Statisticians want numerical measures of the degree to which data support hypotheses. —Hacking

Definition — *inferential model* (IM).

A model for quantifying uncertainty about Z_{n+1} given z^n :

- to every $A \subseteq \mathbb{Z}$, assign a pair of numbers

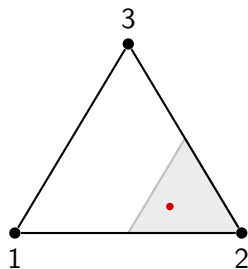
$$\underline{\Pi}_{z^n}(A) \quad \text{and} \quad \overline{\Pi}_{z^n}(A)$$

- data-driven support for and plausibility of “ $Z_{n+1} \in A$ ”

- Distinct $\underline{\Pi}_{z^n} < \overline{\Pi}_{z^n}$ implies *imprecision*
- “Coherent” $(\underline{\Pi}_{z^n}, \overline{\Pi}_{z^n})$ corresponds to a *set of probabilities*
- Precision means $\underline{\Pi}_{z^n} \equiv \overline{\Pi}_{z^n}$, just one prob Π_{z^n}

Manski's Law of Decreasing Credibility: *The credibility of inference decreases with the strength of assumptions maintained*

- What is *imprecise probability*?
- Very simple example: $\mathbb{Z} = \{1, 2, 3\}$
- Points are prob distributions
- Gray: set of probabilities, " $p(2) \geq 0.5$ "
- Wrt gray model: can only give lower and upper bounds on $P(Z \in \cdot)$
- e.g., $0 \leq P(Z \in \{1, 3\}) \leq 0.5$



- Back to the original problem...
- Reasonable strategy:
 - collect all the plausible data-generating process into a set
 - this is a credal set that describes epistemic uncertainty
 - given data, get UQ via “generalized Bayes”⁴
- Accommodates aleatory & epistemic uncertainty
- But data analysts are often *ignorant* or nearly so:
 - credal set is very large
 - generalized Bayes UQ is very imprecise,⁵ inefficient
- For efficiency’s sake, I’ll construct a different IM...

⁴e.g., Walley (1991), *Statistical Reasoning with Imprecise Probabilities*

⁵This is called the *dilation* problem...

Definition — weak validity.

$$P\{\Pi_{Z^n}(A) \leq \alpha, Z_{n+1} \in A\} \leq \alpha, \quad \alpha \in [0, 1], \quad A \subseteq \mathbb{Z}$$

- This is a type of calibration between Π and P
- Gambling interpretation:
 - suppose my UQ Π isn't weakly valid
 - then $P\{\Pi_{Z^n}(A) \leq \alpha, Z_{n+1} \in A\} > \alpha$ for some (α, A)
 - define the gamble

$$g_{Z^n}(Z_{n+1}) = \{\alpha - 1(Z_{n+1} \in A)\} 1\{\Pi_{Z^n}(A) \leq \alpha\}$$

- gamble is acceptable to me, $\Pi_{Z^n}\{g_{Z^n}(Z_{n+1})\} \geq 0$
- but I'd tend to lose money, $P\{g_{Z^n}(Z_{n+1})\} < 0$

- What kind of $z^n \mapsto \Pi_{z^n}$ satisfies weak validity?
- Apply iterated expectation:

$$\begin{aligned} P\{\Pi_{Z^n}(A) \leq \alpha, Z_{n+1} \in A\} \\ = E[P(Z_{n+1} \in A \mid Z^n) 1\{\Pi_{Z^n}(A) \leq \alpha\}] \end{aligned}$$

- Apparently weak validity only holds if

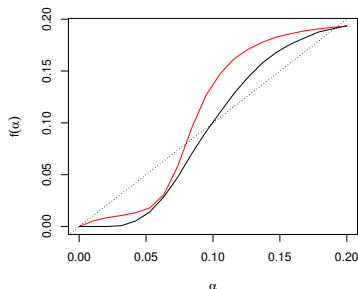
$$\Pi_{Z^n}(A) \geq P(Z_{n+1} \in A \mid Z^n) \quad \text{for all } A$$

- The above condition implies
 - $\Pi_{Z^n}(\cdot) = P(Z_{n+1} \in \cdot \mid Z^n)$, which is out of reach, or
 - Π_{Z^n} is *imprecise*, i.e., an upper probability $\bar{\Pi}_{Z^n}$

- Illustration:

- $n = 5$, $P = \text{Unif}(-5, 5)$, $A = [3, 5]$
- a parametric and nonparametric Bayes predictive distribution
- $f(\alpha) = P\{\Pi_{Z^n}(A) \leq \alpha, Z_{n+1} \in A\}$

- Clearly, weak validity fails



- Prediction confidence sets $\{C_\alpha(z^n) : \alpha \in [0, 1]\}$ satisfy

$$P\{C_\alpha(Z^n) \ni Z_{n+1}\} \geq 1 - \alpha, \quad \alpha \in [0, 1]$$

- Common strategy: flip confidence sets into probabilistic UQ
- e.g., confidence distributions
- But doing so leads to UQ that fails to be weakly valid
- Reliable UQ derived from confidence sets?

- An upper probability $\overline{\Pi}$ is a *possibility measure*⁶ if

$$\overline{\Pi}(A) = \sup_{z \in A} \pi(z), \quad A \subseteq \mathbb{Z}$$

for a function $\pi : \mathbb{Z} \rightarrow [0, 1]$ with $\sup_z \pi(z) = 1$

- π is to $\overline{\Pi}$ what a PDF is to probability, just different calculus
- Simple characterization of confidence to reliable UQ...

Proposition.

For every “nested+” family of confidence sets $\{C_\alpha(z^n) : \alpha \in [0, 1]\}$, there exists a valid possibilistic IM such that

$$z_{n+1} \in C_\alpha(z^n) \iff \pi_{z^n}(z_{n+1}) > \alpha$$

⁶Shafer 1976, Dubois & Prade 1988, etc

- So far:
 - weak validity requires imprecision
 - desirable confidence-like properties are closely connected to possibility theory, a special type of imprecision
- Tie up the loose ends?
 - strong validity: “for all A ” is inside vs outside
 - links up with possibility

Definition — strong validity.

$$P\{\Pi_{Z^n}(A) \leq \alpha \text{ for any } A \text{ with } A \ni Z_{n+1}\} \leq \alpha$$

Proposition.

$$\text{strong validity} \iff P\{\Pi_{Z^n}(\{Z_{n+1}\}) \leq \alpha\} \leq \alpha$$

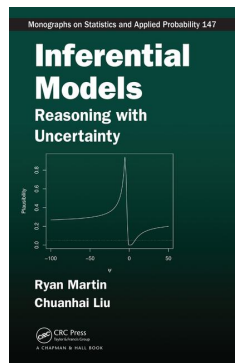
- Latter condition implies $\Pi_{Z^n}(\{Z_{n+1}\})$ can't be bounded away from 1, hence “must be” a possibility measure, say $\bar{\Pi}_{Z^n}$
- If π_{Z^n} is the corresponding contour, then

$$P\{\underbrace{\pi_{Z^n}(Z_{n+1})}_{Z_{n+1} \in C_\alpha(Z^n)} > \alpha\} \geq 1 - \alpha \implies \text{strong validity}$$

- Summary:
 - reliability requires imprecision
 - natural “confidence $\rightarrow$ UQ” is via possibility theory
 - a suitably calibrated possibility measure is weakly and strongly valid, and that's effectively the only way it can be done
- *Question:* How to construct a possibilistic $\bar{\Pi}_{Z^n}$?

Inferential models are one of the original statistical innovations of the 2010s... —Cui & Hannig

- Reliable, data-driven UQ for inference
- Fixes “Fisher’s biggest blunder”
- Original version (as of 2015) →
- New possibilistic perspective:
 - simpler constructions
 - more flexibility
 - ...
- Review article: M. [arXiv:2507.09007](https://arxiv.org/abs/2507.09007)



- A possibility measure $\bar{P}$ is determined by its contour π
- Credal set $\mathcal{C}(\bar{P}) = \{P : P(\cdot) \leq \bar{P}(\cdot)\}$, e.g., gray triangle
- Characterizations of $\mathcal{C}(\bar{P})$ can be given⁷
- Take a fixed P with a unimodal mass/density function p
- Possibility $\bar{P}$ with the smallest $\mathcal{C}(\bar{P})$ that contains P has

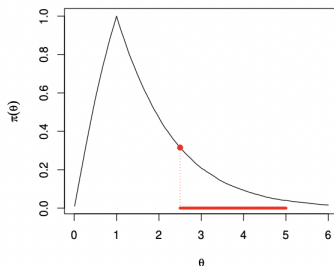
$$\pi(z) = P\{p(Z) \leq p(z)\}, \quad z \in \mathbb{Z}$$

- Called the *probability-to-possibility transform*⁸

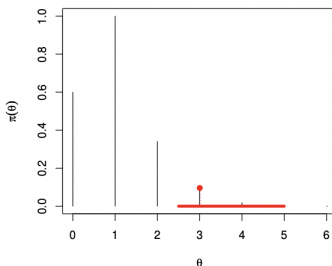
⁷e.g., Couso, Montes, & Gil (2001) and M. arXiv:2501.10585

⁸e.g., Hose's 2022 PhD thesis and Dubois et al (2004)

Possibility theory basics, cont.



(a) $T \sim \text{Gamma}(2, 1)$



(b) $T \sim \text{Bin}(6, 0.2)$

- Plots of the prob-to-poss transform of two different P's
- Can evaluate upper probabilities via optimization
- More general upper expectations via *Choquet integration*⁹

⁹e.g., Troffaes & de Cooman's *Lower Previsions*

- In our prediction problem, however, there's not just one P
- There's an entire convex class $\mathcal{P}$ of probabilities
- So, how can the prob-to-poss transform help?
- *Imprecise probability-to-possibility transform!*¹⁰
 - select a ranking function $\rho : \mathbb{Z} \rightarrow \mathbb{R}$
 - given ρ , define $\bar{\Pi}^\rho$ via the contour

$$\pi^\rho(z) = \sup_{P \in \mathcal{P}} P\{\rho(Z) \leq \rho(z)\}, \quad z \in \mathbb{Z}$$

- With respect to ranking ρ , $\bar{\Pi}^\rho$ is such that $\mathcal{C}(\bar{\Pi}^\rho)$ is the smallest possibilistic credal set that contains $\mathcal{P}$

¹⁰e.g., Hose's 2022 PhD thesis, M. arXiv:2211.14567, & refs therein

- Roughly, a possibilistic IM is just a data-driven instantiation of this imprecise prob-to-poss transform¹¹
- In the context of statistical inference with parametric models, the IM construction can be greatly simplified¹²
- For ML problems, we need the construction's full power
- Let $\rho : \mathbb{Z}^n \times \mathbb{Z} \rightarrow \mathbb{R}$ be a ranking function, large values of $\rho(z^n, z_{n+1})$ means z_{n+1} *conforms to* z^n
- Naive possibilistic IM contour:

$$\pi_{z^n}(z_{n+1}) = \sup_{P \in \mathcal{P}} P\{\rho(Z^n, Z_{n+1}) \leq \rho(z^n, z_{n+1})\}$$

¹¹M. arXiv:2211.14567

¹²M. arXiv:2507.09007

Theorem.

Possibilistic IM $z^n \mapsto \overline{\Pi}_{z^n}$ is strongly valid, hence reliable UQ

- OK, fine, but isn't this $\overline{\Pi}_{z^n}$ impossible to compute?!?!
- *No!* — Solvable problems have structure that allows for dimension reduction that simplifies computation
- This is how IMs for parametric models are “greatly simplified”
- Apply the *Principle of Minimum Complexity*¹³
- Takes different forms depending on the problem, but often boils down to *conditioning*

¹³M. arXiv:2211.14567

- Let's be more careful about the setup...
 - Z^n is observable, interest in Z_{n+1} — no covariates
 - assume exchangeable with unknown dist'n P
 - set of probabilities, $\mathcal{P} = \{\text{exchangeable } P\}$
- Let $\rho : \mathbb{Z}^n \times \mathbb{Z} \rightarrow \mathbb{R}$ be a measure of *conformity*
 - e.g., $\rho(z^n, z_{n+1}) = -d(\hat{z}_{z^n}, z_{n+1})$
 - assume ρ is invariant to permutations of its first argument
- Important/basic facts from your math-stat course:
 - conditional distribution of data, given a sufficient statistic, doesn't depend on any unknown parameters
 - in our nonparametric, exchangeable setting, the unordered data values $\{z_1, \dots, z_n, z_{n+1}\}$ form a sufficient statistic

- *Principle of Minimal Complexity* suggests conditioning on the unordered set of values¹⁴ $\{z_1, \dots, z_n, z_{n+1}\}$

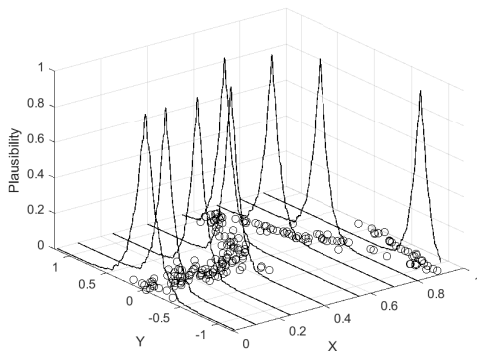
$$\begin{aligned}\pi_{z^n}(z_{n+1}) &= \sup_{P \in \mathcal{P}} P[\rho(Z^n, Z_{n+1}) \leq \rho(z^n, z_{n+1}) \mid \{z^n, z_{n+1}\}] \\ \text{sufficiency} \rightarrow &= \frac{1}{(n+1)!} \sum_{\sigma} 1\{\rho(z^{\sigma(1:n)}, z_{\sigma(n+1)}) \leq \rho(z^n, z_{n+1})\} \\ \text{invariance} \rightarrow &= \frac{1}{n+1} \underbrace{\sum_{i=1}^{n+1} 1\{\rho(z_{-i}^{n+1}, z_i) \leq \rho(z^n, z_{n+1})\}}_{\text{conformal prediction "transducer"}}$$

- Take-aways:
 - IM can be computed
 - strongly valid, fully conditional & coherent possibilistic UQ
 - agrees with conformal prediction in certain ways

¹⁴Faulkenberry (*JASA* 1973), Hoff (*Bernoulli* 2023)

Possibilistic IM prediction, cont.

- For feature-label pairs $Z = (X, Y)$, IM produces possibility contour functions for Y at each $X = x$



- Summary:
 - standard possibilistic IM via prob-to-poss transform
 - *Principle of Minimum Complexity* $\rightarrow$ condition on suff stat
 - unknown P disappears, conformal prediction falls out
- Vovk was right about IMs and conformal prediction!
- But this is more than rediscovering conformal prediction:
 - more than confidence intervals — rigorous, general,¹⁵ fully conditional, and provably reliable UQ
 - new way to view extensions of conformal prediction beyond the standard exchangeable setting
 - e.g., same conditioning argument but allowing for covariate shift returns “weighted conformal prediction”¹⁶

¹⁵e.g., decision theory via Choquet int (M. et al, arXiv:2112.13247)

¹⁶Tibshirani et al (*NeurIPS* 2019)

Two references



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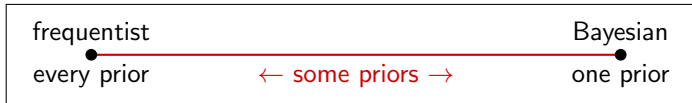
Possibilistic inferential models: a review

[Ryan Martin](#) 

Received 30 Jul 2025, Accepted 02 Dec 2025, Accepted author version posted online: 05 Jan 2026

Conclusion

- Imprecise probability is powerful, and necessary
- Possibilistic IMs can do everything (?) we want
- Probability-to-possibility transform is the workhorse
- Here I showed that conformal prediction is a consequence
- More IM–ML developments are *possible*
- Beyond Bayes and frequentist...



arXiv:2211.14567

The end — *Thanks!*



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

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